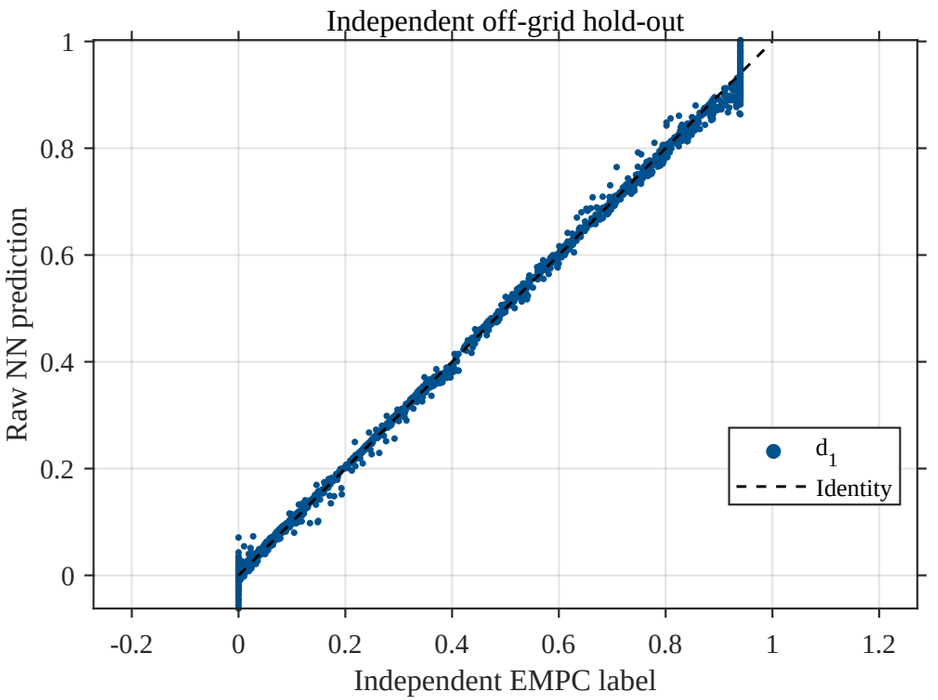
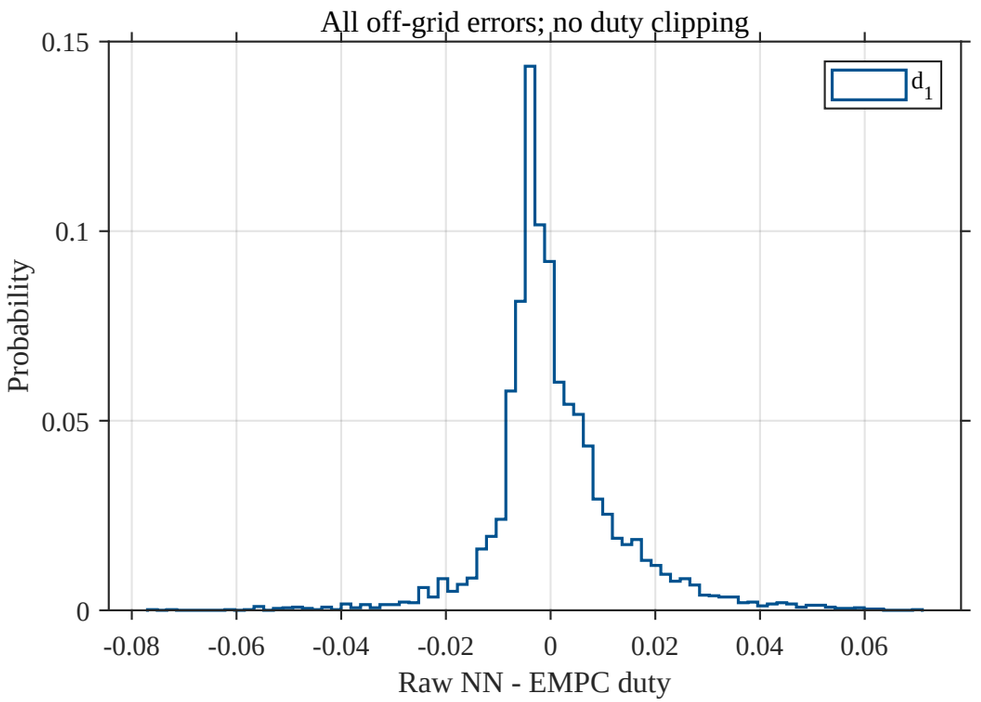
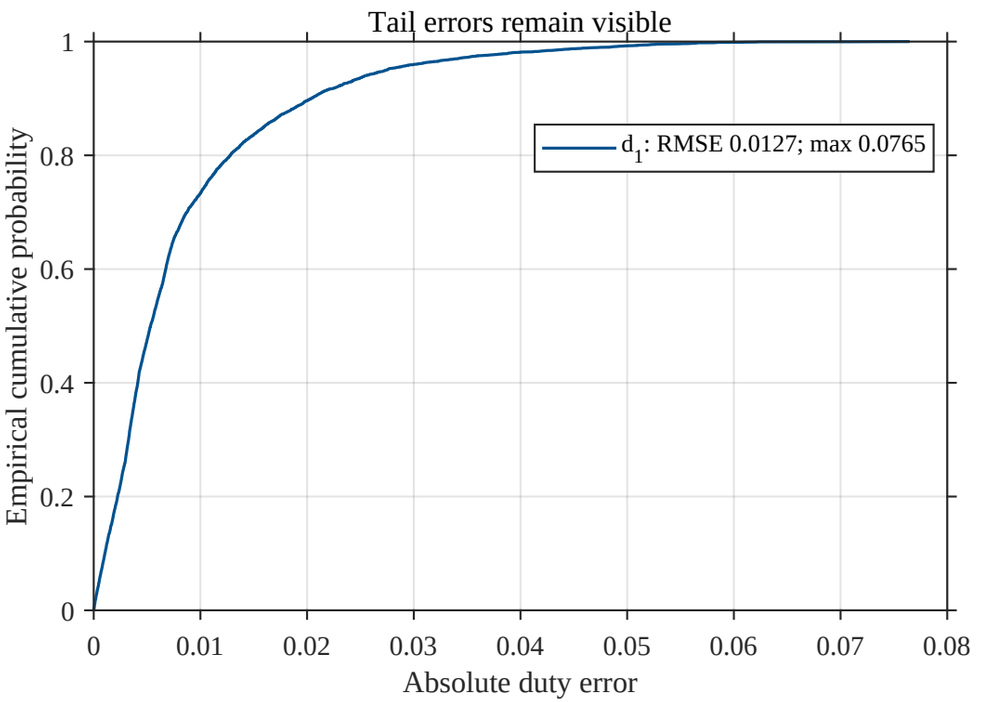

Actual history; goal is not assumed reached
Training
Validation
Goal
Selected update
Goal
MSE (sum across outputs)
Gradient updates
×10^4


Independent off-grid hold-out
Raw NN prediction
Independent EMPC label
d1
Identity


All off-grid errors; no duty clipping
d1
Probability
Raw NN - EMPC duty


Tail errors remain visible
d1: RMSE 0.0127; max 0.0765
Empirical cumulative probability
Absolute duty error